PRABAL KHARE

ROBOTICS SOFTWARE ENGINEER | ROBOTICS ENGINEER | AUTONOMY ENGINEER

Brooklyn, New York | (347) 543 3082 | pk3391@nyu.edu

prabalkhare.com | linkedin.com/in/prabalk | github.com/00PrabalK00

EDUCATION

New York University, Tandon School of Engineering **Expected May 2028**

Master of Science in Mechatronics and Robotics | Brooklyn, New York

Vellore Institute of Technology Chennai **July 2026**

Bachelor of Technology in Computer Engineering, Robotics and AI Specialization | Chennai, India

TECHNICAL SKILLS

Programming: Python, C++, JavaScript and MATLAB
Robotics and autonomy: ROS 2, Nav2, SLAM, tf2, localization, path planning, AMRs, LiDAR, PGV, URDF, Gazebo, MAVROS
Controls and embedded: CANopen, CiA 402, PID control, sensor fusion, Raspberry Pi, Arduino, Pixhawk, PCB design
Learning and robot policies: PyTorch, LeRobot, DINOv2, flow matching, CUDA graphs, Jetson Orin Nano
Vision and tools: OpenCV, YOLOv8, U-Net, COLMAP, FEniCSx, PyQt6, Node-RED, PostgreSQL, Git, Docker, Linux
Spoken languages: English, Hindi, Thai, Marathi and German (basic)

EXPERIENCE

Robotics Software Intern | *Next Robotics Lab Co., Ltd.* **January 2026 to July 2026**

Pathum Thani, Thailand

- Architected a reusable ROS 2 autonomy stack for the 300 kg SMR300 AMR, integrating LiDAR, PGV, Nav2, an AI camera, CANopen and CiA 402 while replacing the company's prior ROS 1 architecture.
- Validated navigation through 150 shelf docking trials and 150 point to point runs, achieving 97% docking success, 2 cm shelf docking error and a 3 cm mean point to point mismatch, down from 5 cm.
- Implemented reflective marker recognition for two and four legged shelves using LiDAR clustering, center checks, automatic repositioning, sector safety controls, safe aborts and alerts; diagnosed failures caused by sunlight reflections at open factory doors.
- Built an operator platform for mapping, map editing, URDF and sensor configuration, Nav2 parameters, mission control, monitoring and deployment validation, plus a Node-RED dashboard evaluated by five operators and two engineers.

Resource Head and Systems Department Member | *Dreadnought Robotics* **August 2023 to June 2025**

Chennai, India | Resource Head, August 2024 to June 2025 | Systems Member, August 2023 to August 2024

- Managed a $15K+ budget, inventory and procurement decisions for 120 members across four departments, comparing alternatives for cost, availability and system fit.
- Designed electrical systems for reliable operation within a sealed AUV hull and developed ROS, MAVROS and MAVLink control software. The team placed second at TAC Challenge Norway 2024.

Hardware Engineering Intern | *Theta Sound* **January 2025 to October 2025**

Remote

- Produced PCB layouts and simulated prototypes for Qualcomm based smart earphones, modeling board footprint, power and battery life; led a KiCad PCB design workshop for 50 attendees.

AI Software Development Intern | *Drone Entry* **June 2025 to August 2025**

Thailand

- Built OpenDroneKit, a PyQt6 structural inspection workflow over approximately 40,000 defect images with YOLOv8 detection and U-Net crack segmentation, mapping cracks onto COLMAP reconstructions and FEniCSx simulations to estimate propagation.

R&D Summer Intern, Robotics and Drones | *Drone Academy* **May 2025 to June 2025**

Thailand

- Designed and fabricated three PCBs (animatronic head, Raspberry Pi 4 I/O, RC car kit); ran text to speech and a 2B parameter language model locally on an 8 GB Raspberry Pi 4.

PROJECTS

Wren, Visuomotor Policy for Low-Cost Robot Arms | *PyTorch, LeRobot, DINOv2, Jetson Orin Nano* **2026**

- Built a 61M-parameter flow-matching policy for the SO-101 arm (DINOv2 encoder, DiT action expert), trained in about 6 hours on an 8 GB laptop GPU; training-time real-time chunking cut chunk-switch disagreement from 1.7° to 0.7°.
- A CUDA graph cut Jetson model time from 288 to 83 ms; 7/10 real-arm successes vs ACT's 5/9, not yet statistically significant.

Ripple, Recovery Agent for Stalled Nav2 Robots | *ROS 2, Nav2, PostgreSQL* **2025 to Present**

- Diagnoses Nav2 stalls and recovers within a configurable budget, escalating to a human for physical blockages and verifying outcomes against live robot state; 13/13 live recovery runs and 9/9 on a second robot profile in Gazebo, not yet run on physical hardware.

Kurat, Autonomous Conversational Robot | *ROS 2, PyTorch, YOLO, STT and TTS* **2025 to Present**

- Three brain perception, cognition and action architecture on a RealSense D435i and Jetson Orin Nano with YOLOv8, Whisper and Nav2.

PATENTS

Lead Inventor on four published Indian patent applications | Applicant: VIT Chennai

Container for Dispensing Medications Oct 2026 · 202641121625
A System for a Companion Robot Oct 2026 · 202641114580
Self-Calibrating Vision System and Method Thereof 2026 · 202641064821
AI-Enabled Search-and-Rescue Robot 2026 · 202641035669